

ENGE702 Engineering Mathematics III

Tutorial 2

1. Consider the differential equation

$$y' - y = e^x.$$

- (a) Sketch the slope field for this differential equation. Plot on integer values of x and y between (and including) -2 and 2 .

- (b) Sketch the solution to the initial value problem

$$y' - y = e^x, \quad y(-2) = 0$$

on your slope field.

- (c) Use Euler's method to estimate $y(2)$, given $y(-2) = 0$. Use a step size of $h = 1$.

} see next page.

$$y(2) \approx 7.27$$

2. Find the general solution to the following ODEs (note they are all separable):

(a) $y' = 2(y - 1)$ $y = Ce^{2x} + 1$

(c) $y' + xy = x$ $y = Ce^{-\frac{x^2}{2}} + 1$

(b) $y' - 4y = 0$ $y = Ce^{4x}$

(d) $xy' = x^3y$ $y = Ce^{\frac{x^3}{3}}$

3. Find the general solution to the following ODEs:

(a) $y' = x + y$ $y = Ce^x - x - 1$

(c) $xy' + 4y = 2x$ $y = Cx^{-4} + \frac{2x}{5}$

(b) $y' + 2y = x^2 + 4x + 7$ $y = Ce^{-2x} + \frac{x^2}{2} + \frac{3x}{2} + \frac{11}{4}$

(d) $xy' - 4y = x^5e^x$ $y = Cx^4 + e^x x^4$

4. Solve the following initial value problems:

(a) $y' = yx^2$ $y(0) = 1$ $y = e^{\frac{x^3}{3}}$

(b) $xy' - xy^2 = 0$ $y(0) = 2$ $y = \frac{1}{\frac{1}{2} - x}$

(c) $y' - 2y = e^x$ $y(0) = 0$ $y = -e^x + e^{2x}$

(d) $xy' + y = x^2$ $y(1) = 0$ $y = \frac{1}{3}(x^2 - \frac{1}{x})$

5. Water flowing through a hole in a tank obeys Torricelli's law, which says

$$v(t) = 0.6\sqrt{2gh(t)},$$

where $v(t)$ is the velocity of the stream of water, $h(t)$ is the height of water above the hole, and $g = 9.81m/s^2$ is the gravitational constant.

Suppose we have a tank that has a diameter of $1m$, a hole (at the bottom) with diameter $2cm$, and an initial height of $3m$.

Set up a model and solve to find a function describing the height of the water in the tank as a function of time.

$$\frac{dh}{dt} = -0.00106\sqrt{h}$$

$$h = (\sqrt{3} - 0.000533t)^2$$

6. Suppose we have an electric circuit with a resistor of 1Ω , and inductor of $1H$ and an EMF given by $E(t) = \sin(t)$. Suppose at time $t = 0$ the current is $I = 0$. Find an expression for the current in the circuit as a function of time.

$$I = \frac{1}{2}e^{-t} + \frac{1}{2}\sin(t) - \frac{1}{2}\cos(t)$$

$$V = -126(10t - \frac{t^2}{2}) + 5000$$

7. A particular machine component has a useful life of 10 years. Its value depreciates at a rate proportional to the life of the product remaining. Suppose it is purchased for \$5000, and one year later its value is \$3800. Find a function that represents the value of the component as a function of time.

8. Finished all these? Work through problem sets 1.3 and 1.5 in the textbook

Computer Part

Even though tutorials are not run in computer labs, during weeks 1 - 6 of the semester, you should work through the section 'Solving Ordinary Differential Equations With MATLAB' (under 'Computational Mathematics') in the self paced courses on the mathworks website, as 'homework'.

If you have never used matlab before, it would be a good idea to work through the section 'MATLAB Onramp' first. Details of how to access these are as follows:

AUT has a licence with the matlab creators that allows you to access an array of online training material, and also download the latest version of matlab to your personal computer (if you wish - you don't need to do this for this course if you don't want to, you can access matlab on the computers at AUT).

To access these resources you will need to create an account with mathworks (the company that produces matlab).

You can do this by visiting mathworks.com (or google 'matlab'). You can create an account by clicking the 'login' icon on the top right. **It is important you use your AUT email address to create the account, or you will not be able to access the material.**

From your account page, click 'Self-Paced Courses'.

1) a,b) something like: $\curvearrowright$ sketched soln

